MEETING SYNTHESIS AND NEXT STEPS

Final Session Notes

At the end of the second day, we discussed the main topics that had arisen over the course of the meeting and generated the following list:

(1) Communication tools

a. We need something more akin to the ISOGEOCHEM list-serv for community-wide announcements. Everyone (including Amy!) agrees that the current approach (i.e., Amy maintaining a master email list) is not tenable in the long run.
b. Some concerns were raised regarding the amount of 'chatter' that occurs on ISOGEOCHEM, but this seems to be endemic to list-servs.
c. There is also an independent isoorbi Slack workspace largely maintained by Caj Neubauer and Seb Kopf, but most of the workshop attendees weren't aware of it prior to Seb's demo. People subscribed to this workspace do use it to announce newly published papers and a fairly regular (monthly?) 'coffee hour' that is used for discussions and/or science presentations. Note, however, that this workspace uses the free version of Slack, so messages and files older than 90 days are inaccessible.

(2) Data repositories

a. **Open Science Data Management: always, always, always upload your .raw files (and at least a READ ME text file) to a data repository** so that they are available to the community. This is good practice for open science. It also provides a resource for groups who are just getting started (i.e., they can look at data like what they might soon be generating and/or play with the data using IsoX) and for our future selves (i.e., who knows what data we might be able to 'mine' after-the-fact).
b. There was some discussion regarding which data repositories would be "best" for storing Orbitrap-IRMS .raw files (which can be extremely large), but ultimately we agreed that individuals should upload to the places that make the most sense given the analytes, applications, etc. and just be sure to include the DOI pointer to the raw data.
c. Note that this discussion was specifically about .raw files, which typically aren't included in Supplementary Information.
    i. **Possible working group discussion**: How to best document metadata associated with Orbitrap-IRMS data? What metadata parameters should be included?

(3) Best practices for reporting methods and data processing

a. It was evident from the talks and posters that many of us are independently making comparable discoveries regarding how different operating / instrument parameters affect our analyses—particularly the accuracy and precision of our isotope ratios.

b. Some parameters, like AGC target, resolution, and quadrupole (i.e., mass window) settings, are obvious things to report in a methods section. However, other parameters are measurement-specific (e.g., maximum injection time, collision energy) or platform-specific (e.g., spray voltages and other ESI settings). The agreed-upon default is to report as much as possible for external reproducibility (Supplements are good for this!). That said, groups working on common platforms (e.g., ESI versus GC), common types of experiments (e.g., direct infusion versus ‘reservoir’ or ‘peak capture’), and potentially even common analytes (e.g., oxyanions, PAHs) should also discuss amongst themselves.
c. Data processing is another realm where we need to be crystal-clear in our manuscripts about what we did to:
    i. Extract the data from our .raw files.
        1. Most people are probably using IsoX now, so this means reporting the information within the isotopologues.tsv files and any associated calculations done using R scripts.
        2. Ideally this means one includes those scripts and .tsv files with the data bundle.
    ii. Convert the extracted values to isotope ratios. For example:
        1. Were the ratios calculated from NL scores (i.e., absolute signal intensities) or from ion counts? If the latter, how were the signal intensities converted to counts?
        2. Are the ratios a summation of all ion counts over all scans, or is the reported ratio an average of scan-by-scan ratios?
    iii. Calculate the reported uncertainties.
    iv. Identify and cull ‘unusable’ data.
        1. What metrics were used to decide whether a scan, acquisition, etc. was ‘acceptable’ or not?
d. There is quite a bit of work being done by Seb Kopf and colleagues in the data processing and data visualization realm. Seb was kind enough to provide a brief overview of some of the work he and others have been doing, which can be found at https://www.isoverse.org.

(4) Education / Establishing an agreed-upon lexicon

a. **For future discussion:** Based on our collective experiences, what are the most important things to teach newcomers to this field?
    i. Note that, for some students, Orbitrap-IRMS may be their first exposure to isotope ratio mass spectrometry!
b. The community would benefit from using an agreed-upon (and well-defined) nomenclature for describing experimental parameters, data processing / analysis, and reporting uncertainties.
    i. For example: What does it mean to calculate isotope ratios from a “block” (as used in some ESI-Orbitrap papers)? Does it consist of a certain number of scans? Is it an average over all scans, or a sum of counts for each isotopologue?

   ii. What do we call the different types of uncertainties associated with Orbitrap analyses? Should we be comparing to shot noise limits or Allen-Werle variance? Both?
   iii. Some ideas for nomenclature were presented in the Eiler et al. (2016) IJMS paper, but these have not been (and need not be) universally adopted within the community.
  c. **ACTION ITEM:** Compiling a 'dictionary' based on the array of current terms and usages would be incredibly useful for the community.

(5) Standards and Standardization
  a. Ring Tests
   i. At the highest-level, the community agreed that—at least among groups analyzing comparable species (e.g., amino acids, nitrates, PAHs) if not among groups using the same platforms (e.g., ESI, GC)—common standards with known isotopic compositions should be circulated among and analyzed by those labs. This is one of the 'best practices' within isotope geochemistry that should be adopted by our community.
   ii. This kind of activity will almost surely have to be self-organized in terms of coordinating between individual labs, but it could be something we kick off (i.e., everyone using ESI-Orbitrap-IRMS takes home a split of analyte X) at next year's meeting.
  b. Position-Specific Isotope Effects
   i. Standards with site-specific labels need to be created and their isotopic compositions independently verified. These are 'niche' compounds because they tend to be tailored to a very specific scientific investigation (and, hence, analyte), and the work can be considerably technically challenging.
  c. How to standardize when pure 'standards' of your analyte don't exist
   i. PFAS groups (among others) have encountered this issue. The question at hand is essentially how well can we standardize our isotope ratios if we need to use a compound other than our analyte as a standard? Some of the approaches tested thus far include using a compound that has a common moiety and/or can be fragmented to create the same (or a functionally similar) fragment to the analyte of interest. Success varies.
  d. Clumped-Isotope Effects
   i. Stochastic reference frames: Our community's analytes of interest by and large cannot be thermally driven to stochastic distributions (unlike the clumped carbonate and methane communities, for example, which uses heated gases). Other clumped isotope communities (e.g., ethane, N2O) cannot heat their standards to stochastic distributions and thus are constrained to work within relative reference frames (i.e., reported with respect to a lab standard). What are our options?
    1. Although some multiply-substituted compounds can be purchased commercially, the available isotopologues are

extremely limited (e.g., only commercial options could be unsubstituted, singly substituted, and completely substituted, and only for a single element).

2. Like PSIA, standardization may require boutique syntheses that necessitate collaboration with our synthetic organic chemistry colleagues; however, this is almost surely not a panacea. There will be limitations.

ii. What if I want a standard that has both PSIA + clumping…?

(6) Collaborations

a. In general, it will be a good idea for us to seek out colleagues who are working on similar types of analyses, similar isotopic systems, and similar instrument platforms. However, we need to avoid the temptation to completely stovepipe ourselves into super-small subgroups and risk losing sight of the larger whole. As a community, we are young enough that we will continue to learn a lot from one another for quite a while. It will be important to maintain community-wide discussions, science exchanges, etc. in addition to more specialized discourse.
b. See Next Steps section below.

(7) Post-Workshop To-Do List

a. Amy to send out a follow-up Google Forms ‘survey’ for feedback on this meeting and to solicit ideas for future meetings (topics, format, location, timing, etc.).
b. Amy to send out material from this meeting (as well as a truncated Google Forms survey) to the larger Orbitrap-IRMS email list.
c. Issaku to investigate possibilities for hosting an Orbitrap-IRMS community list-serv through the University of Utah.

Next Steps:

Several attendees suggested the following large-scale pursuits that the community could coalesce around for long-term, collaborative development, advancement of the field, and to address emergent needs.

International Collaborations / “Open Door” Policies / Global ‘Nodes’

- General concerns regarding the high initial and life-cycle costs of Orbitrap-based mass spectrometers in conjunction with limited research budgets led to several suggestions regarding how to maintain momentum:
    - Some subset of active Orbitrap-IRMS labs (could) have an “open door” policy → anyone in the community is welcome to come and use one of these labs’ instruments as part of a formal collaboration (i.e., working on projects and manuscripts together).

    - John Eiler (Caltech) and Huiming Bao (Nanjing) both vocalized their support for this suggestion and welcomed anyone in the audience to come and collaborate.
    - Q: Are there any Orbitrap-IRMS groups with large enough facilities to create a comparable 'node' in Europe?
  - Some universities have Orbitrap-based mass spectrometers in departments outside of the geo/environmental sciences. Often these instruments are found in biology or biomedical sciences as part of metabolomics / proteomics research programs. Instrument use is typically fee-based, but—as we learned from one of our speakers—some such labs may require that we demonstrate that our analyses will not damage their instruments in some way. (Recall that our analytes and instrument setups / methods are atypical compared to the types of measurements and analytes common to these types of research facilities.)

Working Groups

As captured in the summary notes above, there are a number of issues commonly encountered by all Orbitrap-IRMS users (particularly when starting out). These include, but are not limited to, the following:

- Communication / Education / Common Lexicon
- Data Management and Data Processing
- Method Development and Reporting
- Standardization

While we intend to address as many of these as possible through the annual meeting, the community recognized that there is also great value in working collectively yet asynchronously. Within each of the above categories, there are very different issues specific to the ESI (± HPLC) and GC platforms. (This is true for other, less commonly used ionization sources like APPI, APCI, and LS-APGD, although currently those platforms are more niche.) The ESI and GC subcommunities could thus benefit from their own more-focused working groups. Although there are no formal working group 'assignments', we encourage community members with convergent interests to self-organize, outline a plan to address the issue or issues, pursue those tasks, and report out at next year's meeting.

"Grand Challenges"

John Eiler posited two technical challenges that, should we succeed, could enable significant scientific breakthroughs in many of the fields currently represented within the Orbitrap-IRMS community. He suggested that we consider collaborative efforts over the coming years to tackle the following two in particular:

(1) Metrics of performance: Demonstrate the ability to achieve precision of ≤0.1‰ on any isotope ratio measurement.

(2) Abundance sensitivity: Demonstrate the ability to achieve precision of order ~1‰ on any isotope ratio of an analyte present at femtomolar concentrations.

More generally: Set a goal each year that the community (or a subset of the community) works on together and then writes up for publication.

THANK YOU ALL FOR JOINING US THIS SUMMER, AND WE HOPE TO SEE MANY OF YOU AT OUR SECOND ANNUAL MEETING NEXT YEAR!

AMY HOFMANN, ISSAKU KOHL, & ANDREA ERHARDT (ORGANIZERS)

Mark your calendar for the 2nd Annual Orbitrap-IRMS Users Meeting to be held on 11–12 July 2026 in Montréal, Canada before next year's Goldschmidt Conference!